

Baselines (not trained)

persistence, ema_mean and rls_ar: no offline training; rls_ar learns online per patient.

The baselines are not trained: they have no parameters fitted on the training split. They are what every learned model is compared against.

- **persistence**: predicts $dRR[n+1] = 0$ (the next RR repeats the last one). Nothing to fit.
- **ema_mean**: the next RR is an exponential moving average of the window (factor 0.75 per beat); the factor is fixed, not fitted.
- **rls_ar**: an AR(8) model whose weights are learned ONLINE, per patient, beat by beat, by recursive least squares, never on the training split. It starts at zero weights (it predicts 0, like persistence) with covariance 100 I. When each new beat arrives, the error of the previous forecast updates the 8 weights through the RLS gain; the forgetting factor 0.99 weights the past exponentially (about 100 beats of memory). The regressors are the last 8 RR minus the 30-beat window mean, divided by 100; the forecast is clamped to ± 300 ms.
- Its only design choices (order 8, forgetting 0.99, initial covariance, trace cap $1e4$, clamp) are fixed in the ENGINE settings of src/utls.py; changing them would change the rls_pred feature and invalidate every trained run. The stability guards (skip the update when the covariance loses positive definiteness, keep it symmetric, reset it if its trace becomes non-finite) were added on 2026-10-07, after an unguarded RLS diverged after about 3,200 beats on real data.
- Its forecast is also the rls_pred feature and the residual base: every residual model is trained to correct it, so rls_ar's online learning runs inside every model as well.

Validation reference

baseline	validation RMSE, open loop (ms)
persistence	41.70
ema_mean	43.58
rls_ar	37.14

Abbreviations and acronyms used in this document

term	meaning
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
ema_mean	baseline predicting the next RR as an exponential moving average of the window
HRV	heart rate variability
ms	milliseconds
RLS	recursive least squares: an adaptive filter that updates its coefficients at every beat
rls_ar	baseline: adaptive per-patient AR(8) forecast by RLS
rls_pred	feature: RLS AR(8) forecast of the next dRR (ms), clamped to ± 300
RMSE	root mean square error (ms): the main accuracy measure, lower is better
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms